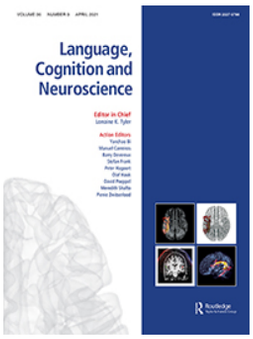

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REGULAR ARTICLE

It's a dotted blue big star: on adjective ordering in a post-nominal language

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ABSTRACT
Referring to a specific object sometimes requires using multiple adjectives. The ordering of the adjectives is assumed to be constrained by universal hierarchies (grammatical or conceptual). It is therefore predicted that different languages will present similar ordering preferences. The ordering in languages where adjectives appear after the noun is further expected to mirror the ordering observed in languages where adjectives appear before the noun. We investigated these predictions in prenominal (English) and postnominal (Hebrew) languages, using three different tasks: production, naturalness rating, and forced-choice. English speakers showed a robust ordering preference. The preferences in Hebrew were significantly weaker. Moreover, for some of the adjective strings, the weakly preferred orders in Hebrew did not mirror the preferred order in English. We argue that constraints on adjective ordering must include additional factors beyond the suggested hierarchies, and discuss this implication in relation to differences between prenominal and postnominal languages.

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Cognitive constraints; word order; subjectivity; reference resolution

1. Introduction

In everyday language, we often have to specify detailed information about the objects that we talk about. This could be done by using several adjectives with the noun that describe the object. When uttering multiple adjectives together, speakers are assumed to have certain preferences for ordering them. For example, in English "the big blue dotted star" is preferred over "the dotted big blue star" or "the blue dotted big star".

These ordering preferences have been the subject of many linguistic and psycholinguistic investigations. Research has tried to understand the strength of the preferences, their manifestation in different languages, and the factors that constrain the ordering (Cinque, 1994; Danks & Glucksberg, 1971; Dixon, 1982; Laenzlinger, 2005; Martin, 1969a, 1969b; Scontras et al., 2017, 2019; Scott, 2002; Svenonius, 2008; Whorf, 1945). In fact, it has been claimed that similar preferences are displayed cross linguistically (Sproat & Shih, 1991). Thus, universal hierarchies of adjective classes have been suggested to determine the relative distance of different adjective classes from the noun. However, the nature of these hierarchies is controversial. Some approaches propose semantic/grammatical hierarchies, which emphasize linguistic and structural representations (Cinque, 1994; Dixon, 1982; Laenzlinger, 2005; Scott, 2002). Others propose perceptual/conceptual hierarchies, which emphasize cognitive factors, such as "absoluteness" and "availability" (Danks & Glucksberg, 1971; Martin, 1969a, 1969b; Scontras et al., 2017, 2019; Sproat & Shih, 1991; Whorf, 1945). Yet, others look at the interaction of many conceptual properties with lexical and phonological ones (Wulff, 2003; Wulff & Gries, 2015).

Regardless of which hypothesis predicts adjective order best, it is widely accepted that adjective ordering preferences are robust (Danks & Glucksberg, 1971; Kemmerer et al., 2009; Martin, 1969a, 1969b). Researchers agree that, for example, when size and color adjectives appear together with the same noun, the size adjective will be produced first in languages like English (Martin, 1969a; Martin & Molfese, 1972; Oller Jr & Sales, 1969) or German (Belke, 2006; Deutsch & Pechmann, 1982) under neutral circumstances (for context manipulation effects, see Danks & Schwenk, 1972; Schwenk & Danks, 1974). Interestingly, a selective impairment of adjective ordering has been reported in patients with left hemisphere damages (Kemmerer, 2000; Kemmerer et al., 2009).

Cross-linguistic variation of multiple adjectives ordering is claimed to be accounted for by variations in the ordering of the noun and its adjectives. That is, differences between prenominal languages, where adjectives appear before nouns, and postnominal language, where adjectives appear after nouns, are assumed. Indeed, linguistic observations reveal differences between prenominal and postnominal languages in adjective order

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preferences (Sproat & Shih, 1991). Based on such observations, it has been argued that postnominal languages, such as Italian (Cinque, 1994), Spanish (Bosque & Picallo, 1996), or Standard Arabic (Fehri, 1999), display order reversal. According to these observations and theoretical accounts, the preferred adjective order in postnominal languages mirrors that of prenominal languages. For example, when size and color adjectives appear together with the same noun, the color adjective will be produced second in prenominal languages, but first in postnominal languages. In both cases, the color adjectives are closer to the noun than the size adjectives. Thus, to describe a car using a size adjective (e.g. big) and a color adjective (e.g. blue), an English speaker will utter the phrase "a big blue car" whereas a Spanish speaker will utter "un coche azul grande" which literally means "a car blue big" (although, see Rosales Jr and Scontras (2019) for empirical data in Spanish; also discussed in the General Discussion).

The current study focussed on another postnominal language, Hebrew. A theoretical account based on linguistic observation argues that adjective ordering preferences in Hebrew mirror those of English (Shlonsky, 2004). However, to the best of our knowledge, this was not tested empirically. In four experiments, we investigated whether Hebrew speakers have a clear preference for a certain adjective ordering. We specifically asked, if such order is observed, whether it mirrors that of English.

In addition to experimentally testing another postnominal language, the current study also aims to examine adjective ordering preferences of more than two adjectives. To date, experimental research on ordering preferences has mainly tested ordering of two adjectives (with many specifically examining size and color; Danks & Glucksberg, 1971; Danks & Schwenk, 1972; Martin, 1969a; Martin & Molfese, 1972; see also Martin (1969b) and Scontras et al. (2017) for other classes of adjectives). Therefore, in three experiments, nouns were modified either by two adjectives or by three adjectives. This allowed us to determine whether robust preferences occur also with three adjectives.

In the two-adjective setting (Experiments 1-3), we used size and color adjectives. The ordering of size and color adjectives is very robust across prenominal languages, such that color is placed closer to the noun than size (for example, "a big blue star"; e.g. Belke, 2006; Danks & Glucksberg, 1971; Martin, 1969a). Indeed, this is the predicted behaviour according to both grammatical and conceptual accounts. For example, perceptual/conceptual theories propose that the adjectives positioned closer to the noun are more conceptually available (defined as an intrinsic (Byrne, 1979), inherent (Whorf, 1945), absolute/accessible (Martin, 1969a), or less subjective (Scontras et al., 2017, 2019) adjective). Color is considered more available than size because size is gradable and therefore requires comparison within-context, whereas color is absolute and not context-dependent, and therefore its comprehension does not require visual or mental comparisons.

In the three-adjective setting (Experiments 1-3), we used size, color and pattern adjectives. To the best of our knowledge, previous research has not provided experimental evidence as to the ordering preferences of these three types of adjectives together. Evidence as described above can help predict the ordering of size and color adjective in the three-adjective setting. That is, size should appear before color also when more adjectives are included in the multi-adjectival phrase. There is also some empirical evidence that English speakers tend to produce pattern adjectives closer to the noun than color adjectives (Fukumura, 2018). Thus, the relative ordering of these adjective classes in English is expected to be size – color – pattern – noun (e.g. "a big blue dotted star").

The adjective classes of size, color and pattern were selected because they were easy to depict visually. This was essential for Experiment 1, which used a production task. In this task, participants were required to produce descriptions of several shapes using two or three adjectives. In the following two experiments, we used close-ended tasks in which participants had to judge the naturalness of multi-adjective descriptions of the same shapes as in the production task (Experiment 2) and choose between two possible descriptions (Experiment 3). The final experiment (Experiment 4) used the same task as Experiment 3 but with several additional adjective classes. If adjective ordering preferences are robust across languages, we expect that the preferred ordering of multiple adjectives in English and Hebrew would be similar, albeit mirrored. Such a preference, if it exists, is expected for both two and three adjective descriptions, and across the three tasks.

2. Experiment 1: production

In the first step, we used a production task due to its implicit nature. In this task, no cues for ordering the adjectives were given from the input, as participants were simply presented with pictures they had to describe. That is, participants were free to choose the ordering of the adjectives without any restrictions. If speakers have a clear preference for adjective ordering with both two and three adjectives, it should present itself in their spontaneous descriptions of the pictures. That is, one specific order is expected to be produced

significantly more than others. Previous studies have shown that certain visual contexts may interfere with speakers' internal ordering preferences. For example, participants produced more Color-Size-Noun orders in English when size was the sole discriminating feature between two different objects than in neutral context (when neither size nor color are needed for contrast; Danks & Schwenk, 1972). Thus, to limit the effects of visual salience of certain visual features, each trial featured a single picture for the participants to describe. Additionally, we examined both Hebrew speakers and English speakers. Any visual interference should have similar effects on both, given the same visual stimuli. Therefore, any differences between English and Hebrew speakers could be attributed to the difference in ordering preferences in their language.

2.1. Method

2.1.1. Participants

Fifty-Three participants participated in this study. Twenty-Seven were native monolingual Hebrew speakers (20-38 years old, Mean = 25) and Twenty-Six were native monolingual English speakers (20-45 years old, Mean = 32.3), all had no history of reading difficulties or any other cognitive impairment. Hebrew speakers were recruited via social media, and volunteered to participate. English speakers were recruited via the "Prolific" internet platform.

2.1.2. Materials

Participants were presented with pictures of a single simple shape (a star, a circle, a triangle, or a square). We used a single item, rather than an array of items, to avoid interference (through visual salience) with the ordering preferences. For the two-adjective setting, the shapes were colored in a solid color (red, blue or green). To provide the means to determine size, we used a frame around each picture, such that there were big shapes, encompassing almost all of the space within the frame (Figure 1A) or small shapes, encompassing little space within the frame (Figure 1B). For the three-adjective setting, we used similar pictures, but a pattern was added, instead of the solid color (dotted or checkered; see Figure 1C and D). There were fourteen items, seven different items for each setting (two or three adjectives). Items were divided into two lists, counterbalanced for the type of the first item (two-adjective item or three-adjective item).

2.1.3. Procedure

Participants were presented with a single item in each trial. Below each item, we included an empty line, where participants were asked to type the description of the item. We instructed them to provide a clear description, such that another person, who cannot see the picture, would be able to draw the shape. We asked for short descriptions that did not include the location of the item within the frame. To ensure that

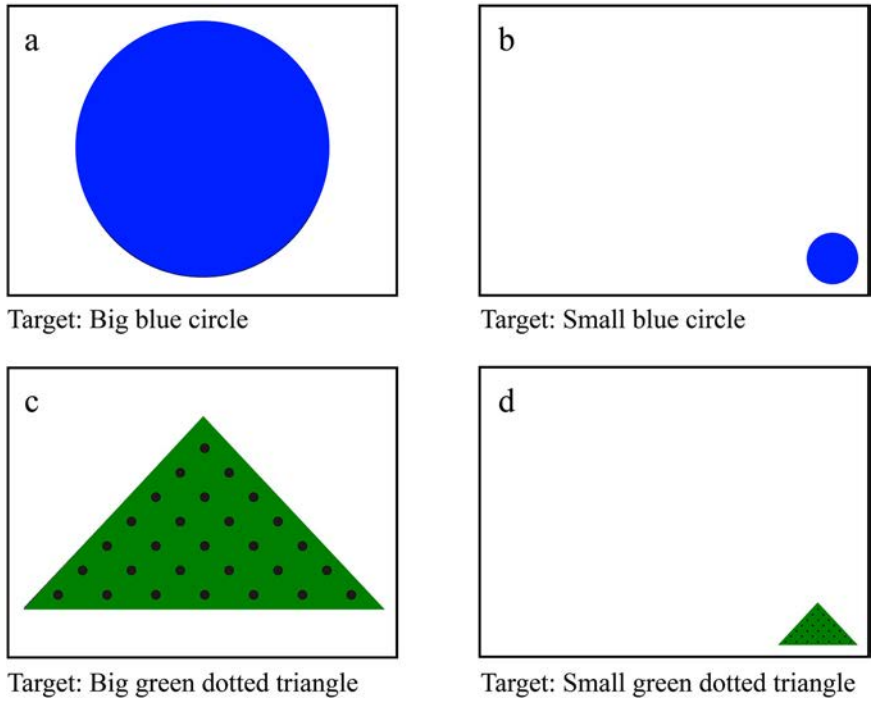

Figure 1. Examples of our experimental stimuli.

participants will not use prepositional phrases for pattern adjectives (such as "with dots"), we provided a word-bank which was shown only in the instruction page. The word bank included the adjectives and the shapes (Red, Big, Green, Star, Blue, Dotted, Checkered, Triangle, Circle, Small, Square) in four different orders. Thus, there were 8 lists in total (word bank order X type of first item). The average time for completing the study was approximately 10 min in both languages.

2.2. Results

For this analysis, we included only full descriptions, with two adjectives (color and size) in the two-adjective setting, and with three adjectives (color, size and pattern) in the three-adjective setting. Moreover, we excluded descriptions with words not included in the word-bank. For Hebrew, some of the productions (18% of the productions in the three-adjective items and 11% of the productions in the two-adjective items) included "and" before the last adjective. These productions were also excluded.[1] At the end, there were 155 productions with two adjectives and 73 productions with three adjectives in English, and 173 productions with two adjectives and 92 productions with three adjectives in Hebrew.

Because one of our aims was to determine whether adjectives ordering preferences in postnominal languages mirror those of prenominal languages, we adjusted adjectives ordering in the productions accordingly. For example, the order Pattern-Color-Size-Noun in English is coupled with its mirror order, Noun-Size-Color-Pattern, in Hebrew. In our coding, we used the first letter of each adjective class, and listed the two mirror orders, the first in English and the second in Hebrew (see Table 1). Thus, the coding for the orders given above is PCS-N/N-SCP. In the comparison between the languages, we considered the preferred order in English as the target order (Size-Color, or SC-N/N-CS, for the two-adjective setting, and Size-Color-Pattern, or SCP-N/N-PCS, for the three-adjective setting).

Table 1. Order coding for the different adjective orders in English and in Hebrew.

Order Code	Spelled out order in English	Spelled out order in Hebrew
SC-N/N-CS[a]	Size-Color-Noun	Noun-Color-Size
CS-N/N-SC	Color-Size-Noun	Noun-Size-Color
SCP-N/N-PCS[a]	Size-Color-Pattern-Noun	Noun-Pattern-Color-Size
SPC-N/N-CPS	Size-Pattern-Color-Noun	Noun-Color-Pattern-Size
PSC-N/N-CSP	Pattern-Size-Color-Noun	Noun-Color-Size-Pattern
PCS-N/N-SCP	Pattern-Color-Size-Noun	Noun-Size-Color-Pattern
CSP-N/N-PSC	Color-Size-Pattern-Noun	Noun-Pattern-Size-Color
CPS-N/N-SPC	Color-Pattern-Size-Noun	Noun-Size-Pattern-Color

[a]This is the target order in the analysis, based on the reported preference in English

2.2.1. Two-adjective setting

English speakers produced the SC-N order 100% of the time, whereas Hebrew speakers produced the same (mirrored) order (N-CS) approximately 71% of the time (Figure 2a). To examine whether Hebrew speakers prefer this order over the other possible order (N-SC), we conducted a binomial test. We set the chance level at 1/2 because speakers could have produced only two possible orders. This test revealed that the N-CS order

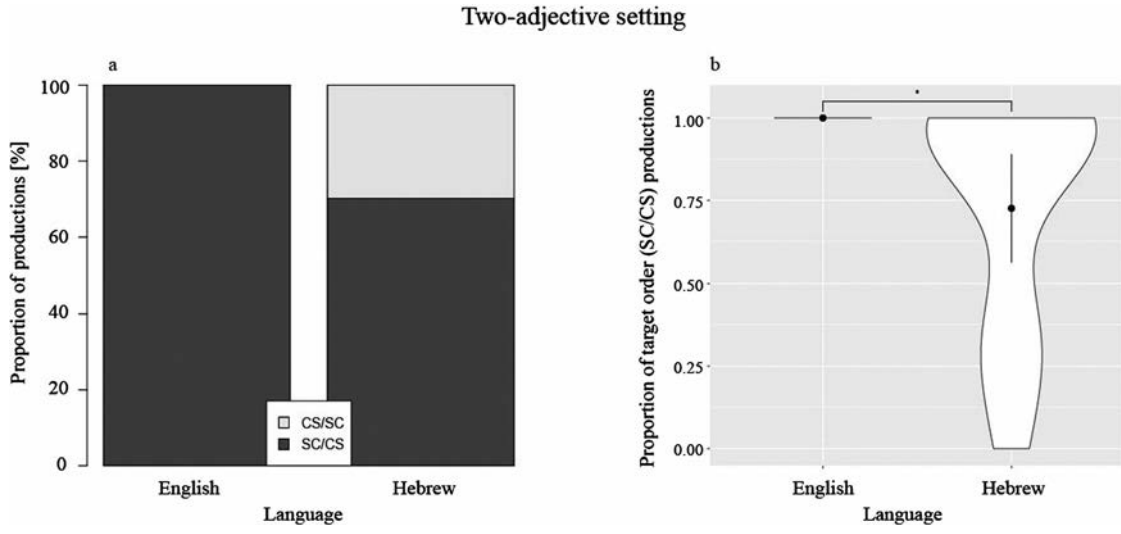

Figure 2. (a) Percentage of productions of each possible adjective orders in the two-adjective setting; (b) Productions of the target order in English (SC-N; N-CS in Hebrew), by language. Error bars represent group 95% confidence intervals and points represent group means. S = Size; C = Color. Note that in English (to the left of the slash) the adjectives precede the noun, and in Hebrew (to the right) the noun precedes the adjectives. * $p < 0.05$.

Table 2. Summary of the mixed-effects logistic regression models in the production task, examining ordering preferences of the target order in two-adjective (SC-N/N-CS) and three-adjective (SCP-N/N-PCS) settings in the English (reference) vs. Hebrew.

Setting	B	SEM	z-value	*P*
2 adjectives: English vs. Hebrew	2.89	1.28	2.25	0.0244 *
3 adjectives: English vs. Hebrew	9.58	3.79	2.52	0.0116*

was produced more often than chance and the N-SC order was produced at below chance level (*p*s < 0.001).

To examine the differences between the two languages, we performed a mixed-effects logistic regression using the lme4 package (Bates et al., 2015) in the R environment (R Core Team, 2015). We modelled the odds of producing the target order (SC-N/N-CS), and tested the differences between the two languages. Thus, our fixed effect was language (Hebrew/English). We also included random intercepts for participants and for items (without random slopes due to the failure of the full model to converge). The model revealed an effect of language (Table 2 and Figure 2b), indicating that the SC-N/N-CS order (Size-Color-Noun in English; Noun-Color-Size in Hebrew) was produced significantly more often in English than in Hebrew.

2.2.2. Three-adjective setting

As expected, most of the productions in English used the SCP-N order (74%). The rest (26%) used the SPC-N order, with a single production in the CPS-N order (Figure 3a). Hebrew speakers produced a wider variety of orders: N-SCP (30.4%), N-CSP (28.3%), N-CPS (25%), N-PCS (the mirror of the target order in English, 14.1%), and N-SPC (2.2%). These descriptive results suggest a clear preference in English, but not in Hebrew.

To verify this statistically, we conducted a binomial test, setting the chance-level at 1/6 because speakers could have produced one of six possible orders. English speakers produced the SCP-N order more often than chance ($p < 0.001$). Rates of SPC-N production did not differ from chance, after correcting for multiple comparisons ($p = 0.028$, not significant after applying Bonferroni adjustment set at 0.0083). Hebrew speakers produced two orders more frequently than chance: N-SCP ($p = 0.0007$) and N-CSP ($p = 0.003$). Production rates of the target order (N-PCS) and the N-CPS order did not differ from chance. All other orders, both in English and in Hebrew, were produced significantly below chance (*p*s < 0.001).

The results of the binomial test offer a first indication that ordering preferences for three adjectives with the classes of size, color, and pattern are different between Hebrew and English. To confirm this, we further performed a mixed-effects logistic regression using the lme4 package (Bates et al., 2015) in the R environment (R Core Team, 2015). We modelled the odds of producing the target order (SCP-N/N-PCS), and tested the differences between the languages. Thus, our fixed effect was language. Random intercepts were included for participants and items (without random slopes due to the failure of the full model to converge). This analysis revealed an effect of language (Table 2 and Figure 3b), indicating that the target order, SCP-N/N-PCS, (Size-Color-Pattern-Noun in English and Noun-

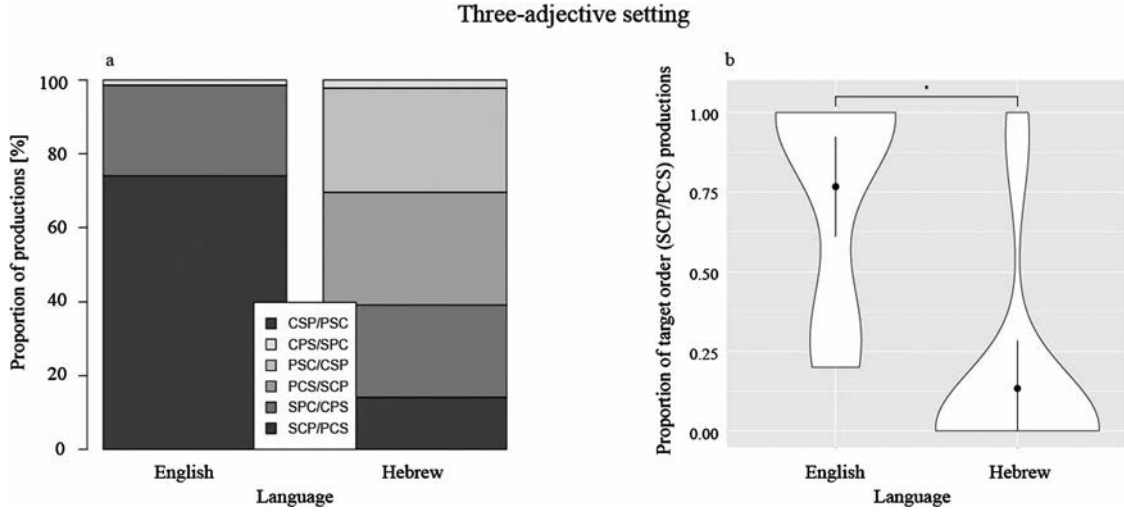

Figure 3. (a) Percentage of productions of each possible adjective order in the three-adjective setting; (b) Productions of the target order in English (SCP-N; N-PCS in Hebrew), by language. Error bars represent group 95% confidence intervals and points represent group means. S = Size; C = Color; *P* = Pattern. Note that in English (to the left of the slash) the adjectives precede the noun, and in Hebrew (to the right) the noun precedes the adjectives. * $p < 0.05$.

Table 3. Mixed effects regression model coefficients for the two-adjective setting in the naturalness rating task, examining differences in naturalness rating between the SC-N/N-CS (reference) and the CS-N/N-SC orders in English (reference) vs. Hebrew.

Contrast	β	SEM	t-value	df	*p*
Language (English): Hebrew	0.43	0.21	48.98	2.11	0.0404*
Order (SC/CS): CS/SC	−0.77	0.09	625.85	−8.96	<0.001***
Language (English): Hebrew X Order (SC/CS): CS/SC	1.36	0.18	303.01	7.68	<0.001***

Note that in English (to the left of the slash) the adjectives precede the noun, and in Hebrew (to the right) the noun precedes the adjectives.

Pattern-Color-Size in Hebrew) was produced significantly more frequently by English speakers than by Hebrew speakers.

This result suggests that English speakers' preference for the target order is significantly stronger than Hebrew speakers'. However, it is possible that, unlike the theoretical assumption, the preferred order in Hebrew does not mirror that of the preferred order in English. As one of our aims was to determine whether the strength of adjective ordering preferences is similar in Hebrew and English, we performed another analysis with different adjective orders. We compared the most produced order in each language (SCP-N in English and N-SCP in Hebrew). This analysis is an ad-hoc analysis based on the observed data, but it allows some insight to the nature of preferences regarding adjective ordering in both languages, beyond a specific order. We applied the same model used in the above regression, testing the difference in the odds of producing the most produced order. This model too revealed an effect of language ($p = 0.015$). That is, even when taking into account different preferred orders in the two languages, the preference in English is substantially stronger than in Hebrew.

2.3. Discussion

In this experiment, participants spontaneously produced phrases with multiple adjectives. We examined the order of size and color adjective in a two-adjective setting (where a shape had 2 visual features) and the order of size, color and pattern adjectives in a three-adjective setting (where a shape had 3 visual features). Our results suggest a significant difference in adjective ordering preferences between English speakers and Hebrew speakers, on both the strength of the preferences (in both two- and three-adjective settings) and the preferred orders themselves (in the three-adjective setting). In both languages, speakers preferred the same mirrored order, Size-Color-Noun in English and Noun-Color-Size in Hebrew, when producing phrases with two adjectives. With three adjectives, the expected Size-Color-Pattern-Noun order was favoured by English-speakers. Hebrew speakers, on the other hand, produced two orders, Noun-Size-Color-Pattern and Noun-Color-Size-Pattern, in rates higher than chance. Notably, ordering preferences of multiple adjectives were much more robust for English speakers than for Hebrew speakers. In fact, for the two-adjective production, English speakers never produced the opposite order, whereas Hebrew speakers produced it almost 30% of the times. With three adjectives, Hebrew speakers produced more distinct orders than English speakers did.

Crucially, neither of the most produced orders (Noun-Size-Color-pattern and Noun-Color-Size-Pattern) in Hebrew mirrored the preferred order in English (i.e. Size-Color-Pattern-Noun). Interestingly, in the two preferred orders in Hebrew, the pattern adjectives were placed the farthest from the noun, unlike in English where they were most frequently placed the closest to the noun. One explanation for this could be denominalization of the pattern adjectives. The reader might have noticed that the English pattern adjectives we used, such as "dotted", are denominal. Some theoretical accounts proposed that denominal adjectives are placed closer to the noun (Bache, 1978, inter alia). However, the Hebrew pattern adjectives are similarly denominal (***nekuda***$_{\text{HEB}}$ for the English word "dot", and *me**nukad***$_{\text{HEB}}$ for the English word "dotted", where changes in the word template are part of the Semitic morphological system of Hebrew). That is, both languages should show a preference to place the denominal pattern closer to the noun. Therefore, this account cannot explain the differences between the languages.

This finding, that multi-adjective phrases have different preferred orders in English and Hebrew, has some critical implications for accounts that argue for mirroring order in postnominal languages (Shlonsky, 2004; Sproat & Shih, 1991). The following experiments were performed in order to confirm this finding in different experimental settings that may allow a more detailed examination of the preference in Hebrew.

3. Experiment 2: naturalness ratings

Experiment 1 showed that Hebrew and English speakers produced different orders when using three adjectives to refer to the same visual objects. In this experiment, we further examined this effect by collecting naturalness ratings of several phrases with different adjective orders. Using naturalness ratings, a task relevant to language

perception, we could determine whether the same ordering preferences go beyond language production. Moreover, in the production task speakers could produce only one order, whereas the naturalness rating task allowed participants to assign high scores for several orders. Thus, this task could reveal whether more than one order is preferred.

3.1. Method

3.1.1. Participants

Forty-nine participants participated in this experiment: Twenty-four were native monolingual Hebrew speakers (20-45 years old, Mean = 30.2) and twenty-five were native monolingual English speakers (20-45 years old, Mean = 31.3). They had no history of reading difficulties or any other cognitive impairment. Hebrew speakers were recruited via social media, and volunteered to participate. English speakers were recruited via "Prolific". The average time for completing the study was approximately 10 min. Data from other 35 English-speaking participants and 5 Hebrew-speaking participants were excluded from analysis, due to low quality responses.[2]

3.1.2. Materials

Fifty-six pictures, similar to those in Experiment 1, were used. The pictures were presented with the maximal description of the shape they depicted. For the two-adjective setting, we used fourteen pictures (Figure 1A above), half of which were described using the SC-N/N-CS order (e.g. "Big blue star") and half were described using the CS-N/N-SC order (e.g. "Blue big star"). For the three-adjective setting, forty-two shapes were used with seven repetitions for each possible order (PSC-N/N-CSP, PCS-N/N-SCP, SCP-N/N-PCS, SPC-N/N-CPS, CSP-N/N-PSC, CPS-N/N-SPC). We also included six catch trials, in which the wrong color, size, pattern or shape were used in the description (e.g. "Big green circle" to describe the picture in Figure 1A). There were eight randomised lists, each beginning with a different adjective-order. To avoid including low quality responses, we excluded participants who gave ratings higher than 1 in two or more catch trials.

3.1.3. Procedure

Participants were asked to rate, on a scale of 1–7, how natural each description was for the picture it was presented with. The rating "1" was used for unnatural descriptions and the rating "7" was used for very natural description.

3.2. Results

Like in Experiment 1, we adjusted the adjective ordering in Hebrew to allow a comparison between the two languages (see Table 1).

3.2.1. The two-adjective setting

As expected, the average ratings for the target SC-N/N-CS order were high in both languages (mean rating = 6.47 and 6.69 in Hebrew and English respectively). The average ratings for the CS-N/N-SC order were 6.36 and 5.27 in Hebrew and English, respectively (Figure 4). To examine the differences in ratings, we conducted a mixed effects linear regression, using the lme4 package (Bates et al., 2015) in the R environment (R Core Team, 2015). We modelled the naturalness rating of each description. Our fixed effects were language (Hebrew/English) and order (SC-N/CS-N). We included random intercepts for participants. Random intercepts for items, as well as random slopes for participants and items, were excluded from the model due to its failure to converge.

This analysis revealed an effect of language (see Table 3), indicating that the ratings in Hebrew were higher than in English. An effect of order was also observed (Table 3), indicating that the SC-N/N-CS order was rated higher than the opposite order across the languages. Finally, there was also a significant

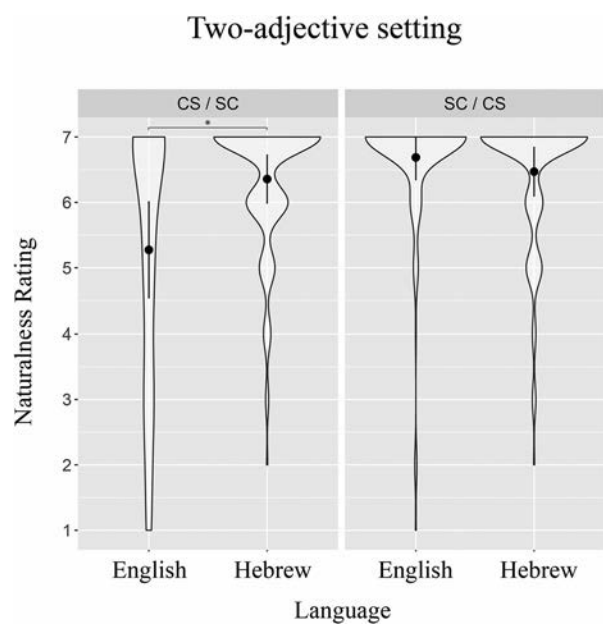

Figure 4. Naturalness ratings in the two-adjective setting; Error bars represent group 95% confidence intervals and points represent group means. S = Size; C = Color. Note that in English (to the left of the slash) the adjectives precede the noun, and in Hebrew (to the right) the noun precedes the adjectives. * $p < 0.05$.

interaction between language and order. Follow-up pairwise comparisons, adjusted by Bonferroni adjustment for multiple comparisons (significance threshold set at 0.0125), revealed that the ratings for the two orders did not differ in Hebrew ($p = 0.481$), but the SC-N order was rated higher than the CS-N order in English ($p < 0.001$, see Table A.1 in supplementary materials A for the detailed pairwise comparisons). In addition, the ratings for the SC-N/N-CS order were similar in the two languages ($p = 0.288$), whereas the ratings were higher for N-SC order in Hebrew than for CS-N order in English ($p < 0.001$).

3.2.2. The three-adjective setting

The average rating for the SCP-N order in English was very high (mean rating = 6.37). Similarly to the production results, the SPC-N order was also rated relatively high (mean rating = 5.83), whereas the average ratings for all other orders were below 4.8 (Figure 5). In Hebrew, the ratings were generally lower, with maximal average rating of 5.48 for the N-CSP order.

We conducted a mixed effects linear regression, using the lme4 package (Bates et al., 2015) in the R environment (R Core Team, 2015), where we modelled those naturalness ratings. Our fixed effects were language (Hebrew/English) and order (SCP-N/SPC-N/PSC-N/PCS-N/CSP-N/CPS-N in English and their mirror images in Hebrew). We included random intercepts for participants and for items. Random slopes for both participants and items were excluded from the model due to its failure to converge. This model revealed effects of order ($ps < 0.025$; Table 4), and a significant interaction between language and order ($ps < 0.001$). No effect of language was observed.

To understand the source of the interaction, we performed follow up pairwise comparisons. We examined corresponding adjective orders across languages, based on mirroring adjustments (see Table 1). The preferred order in English (SCP-N) was rated higher than its corresponding mirror order in Hebrew ($p < 0.001$, Table A.2 in Supplementary materials A for all the details on the follow up pairwise comparisons). No other differences were observed (see Table A.2). This further supports the finding from Experiment 1 that adjective ordering in Hebrew, when using three adjectives, does not mirror that of English. Additional follow-up pairwise comparisons between orders within each language were performed (full results are given in the supplementary materials, Table A.3 and A.4). Of note to our question regarding the strength of the ordering preference, we consider the fact that more comparisons revealed significant differences in English than in Hebrew. Furthermore, the effect sizes, as reflected by the β estimates, were larger in English than in Hebrew.

3.2.3. Exploratory analysis of adjective hierarchies in the three-adjective setting

We conducted further analyses to explore ordering hierarchies in Hebrew and in English. For these analyses, we examined each class of adjective (color, size, pattern) separately. These analyses were conducted in order to

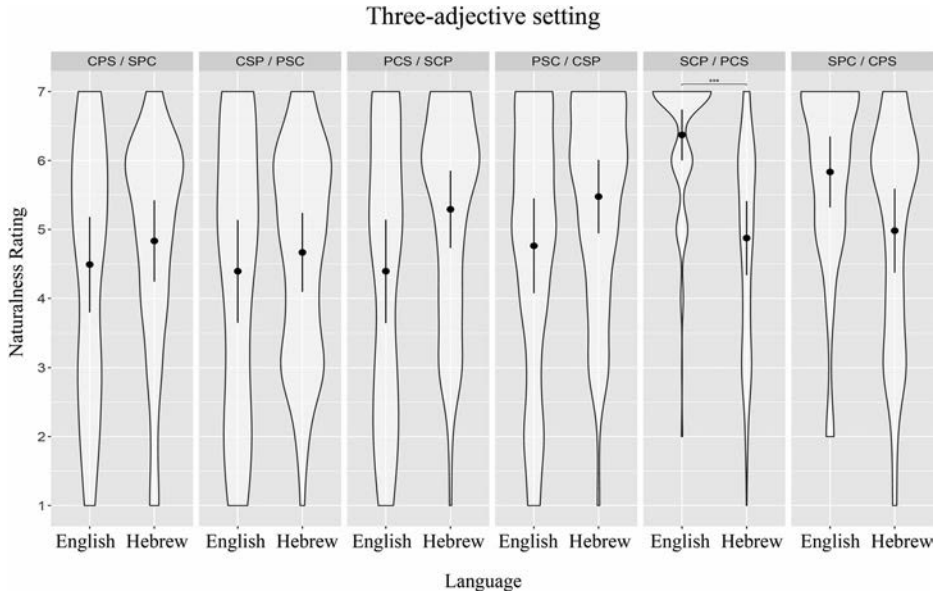

Figure 5. Natural ratings in the three-adjective setting; Points represent group mean, error bars represent 95% confidence intervals. S = Size; C = Color; *P* = Pattern. Note that adjectives in Hebrew (after the slash) follow the noun, whereas in English (before the slash) they precede it. ***$p < 0.001$.

Table 4. Mixed effects regression model coefficients for the three-adjectives setting, examining differences in naturalness rating between SCP-N/N-PCS (reference) and all other orders in English (reference) vs. Hebrew.

Contrast	β	SEM	df	t-value	*p*
Language (English): Hebrew	−0.02	0.30	48.88	−0.07	0.944
Order (SCP/PCS): SPC/CPS	−0.22	0.09	2011.01	−2.28	0.023*
Order (SCP/PCS): CPS/SPC	−0.96	0.09	2011.01	−10.18	<0.001***
Order (SCP/PCS): CSP/PSC	−1.09	0.09	2011.01	−11.57	<0.001***
Order (SCP/PCS): PCS/SCP	−0.78	0.09	2011.01	−8.26	<0.001***
Order (SCP/PCS): PSC/CSP	−0.50	0.09	2011.02	−5.33	0.001***
Language (English): Hebrew X Order (SCP/PCS): SPC/CPS	0.64	0.19	2011.01	3.41	0.001***
Language (English): Hebrew X Order (SCP/PCS): CPS/SPC	1.84	0.19	2011.01	9.74	<0.001***
Language (English): Hebrew X Order (SCP/PCS): CSP/PSC	1.77	0.19	2011.01	9.37	<0.001***
Language (English): Hebrew X Order (SCP/PCS): PCS/SCP	2.39	0.19	2011.01	12.68	<0.001***
Language (English): Hebrew X Order (SCP/PCS): PSC/CSP	2.21	0.19	2011.02	11.71	<0.001***

** Note that in English (to the left of the slash) the adjectives precede the noun, and in Hebrew (to the right) the noun precedes the adjectives.

provide a clearer indication for the ordering preferences of each adjective class relative to the noun. Looking at each class separately, we should be able to make stronger conclusions that go beyond each specific sequence. Thus, we conducted three separate mixed effects linear regressions. In each model, we examined the position of the tested adjective class in relation to the noun. For that, we coded the position of each adjective class based on its distance from the noun, using "1", "2", or "3" (Table 5). For example, for the SPC-N/N-CPS order, we used "3" in the size analysis (because size is positioned the farthest from the noun), but "1" in the color analysis (because color is positioned closest to the noun). In each model, our fixed effects were language (Hebrew/English) and distance from the noun ("1", "2", "3"). Random intercepts of participants and items were included in each model, but random slopes were included only for participants due to the failure of the full model to converge.

3.2.3.1. Size ordering. Our model revealed an effect of distance from the noun when Distance "3" was compared with Distance "1", and an interaction between language and distance in the same comparison (*ps* < 0.001; Table 6). There was no effect for language ($p = 0.943$). To follow-up on the interaction, we conducted 9 pairwise comparisons (using Bonferroni adjustment). We compared the naturalness ratings of the three possible positions ("1" vs."2", "2" vs. "3", and "1" vs. "3") within each language, and the naturalness ratings between languages ("1" in English vs. "1" in Hebrew, "2" in English vs. "2" in Hebrew, and "3" in English vs. "3" in Hebrew) within each position.

In English, size adjectives were preferred in the position farthest from the noun (i.e. marked "3"), as evident by the higher ratings for Distance "3" compared with Distance "2" and Distance "1" (*ps* < 0.001; see Table A.5 in Supplementary materials A for more details). Naturalness ratings for size in Distance "2" and in Distance "1" were not significantly different ($p = 0.148$). Hebrew speakers, on the other hand, did not show a strong preference for the position of size adjectives, as no differences were found in all distance comparisons (*ps* > 0.14). The difference between the languages was also evident in the within-position comparison: we observed a significant difference when comparing Distance "3" in the two languages. That is, orders where the size was farthest from the noun were rated higher in English than in Hebrew ($p < 0.001$).

Table 5. Position coding for the exploratory analysis of adjectives' distance from the noun.

Order	Position of Size adjective	Position of Color adjective	Position of Pattern adjective
SCP/PCS	3	2	1
SPC/CPS	3	1	2
CSP/PSC	2	3	1
CPS/SPC	1	3	2
PSC/CSP	2	1	3
PCS/SCP	1	2	3

Note that in English (to the left of the slash) the adjectives precede the noun, and in Hebrew (to the right) the noun precedes the adjectives.

3.2.3.2. Color ordering. Like in the analysis for size ordering, our model revealed an effect of distance from the noun when Distance "3" was compared with Distance "1" ($p < 0.001$; Table 7), and an interaction

Table 6. Mixed effects regression model coefficients for the position of size adjectives, examining differences in naturalness ratings between descriptions where size was placed at a distance of 1 from the noun (reference) and all other distances, in English (reference) vs. Hebrew.

Contrast	β	SEM	df	t.ratio	*p*
Language (English): Hebrew	−0.02	0.30	48.86	−0.07	0.943
Distance (1): 2	0.07	0.07	2011.02	1.08	0.281
Distance (1): 3	0.76	0.07	2011.01	11.23	<0.001***
Language (English): Hebrew X Distance (1): 2	−0.13	0.14	2011.02	−0.95	0.343
Language (English): Hebrew X Distance (1): 3	−1.79	0.14	2011.01	−13.20	<0.001***

Table 7. Mixed effects regression model coefficients for the position of color adjectives, examining differences in naturalness ratings between descriptions where color was placed at a distance of 1 from the noun (reference) and all other distances, in English (reference) vs. Hebrew.

Contrast	β	SEM	df	t	*p*
Language (English): Hebrew	−0.02	0.30	49.00	−0.07	0.947
Distance (1): 2	−0.03	0.07	2011.00	−0.42	0.678
Distance (1): 3	−0.67	0.07	2011.00	−9.29	<0.001***
Language (English): Hebrew X Distance (1): 2	−0.23	0.14	2011.00	−1.62	0.106
Language (English): Hebrew X Distance (1): 3	0.38	0.14	2011.00	2.61	0.009**

between language and distance in the same comparison ($p = 0.009$). Again, there was no effect for language ($p = 0.947$). To follow up on the interaction, we conducted 9 pairwise comparisons, similar to the size analysis (using Bonferroni adjustment). In English, both the comparisons between "1" and "3" and between "2" and "3" revealed significant differences (*ps* < 0.001, Table A.5 for more details). The comparison between "1" and "2" was not significant. Hebrew speakers displayed similar results to English speakers, with significant differences between distances "1" and "3" and "2" and "3" (*ps* < 0.001), and no difference between "1" and "2". Indeed, no significant effects were observed when comparing Hebrew and English within each position. That is, color adjectives positioned the closest to the noun or in the middle position had higher ratings than when positioned farthest from the noun in both languages. The significant interaction with the null simple effects indicates that the difference in ratings between the distances in English was larger than in Hebrew. This is in line with our finding that the adjective order preferences in English are more robust than those in Hebrew.

3.2.3.3. Pattern ordering. Our model revealed a significant interaction between language and distance from the noun when Distance "3" was compared with Distance "1" ($p < 0.001$, Table 8). No effect of language

Table 8. Mixed effects regression model coefficients for the position of pattern adjectives, examining differences in naturalness ratings between descriptions where pattern was placed at a distance of 1 from the noun (reference) and all other distances, in English (reference) vs. Hebrew.

Contrast	β	SEM	df	t-value	*p*
Language (English): Hebrew	0.021	0.30	48.90	−0.07	0.943
Distance (1): 2	−0.04	0.07	2011.00	−0.58	0.565
Distance (1): 3	−0.10	0.07	2011.00	−1.31	0.19
Language (English): Hebrew X Distance (1): 2	0.36	0.14	2011.00	2.47	0.014
Language (English): Hebrew X Distance (1): 3	1.42	0.14	2011.00	9.81	<0.001***

($p = 0.943$) nor distance (*ps* > 0.18) were observed. To follow up on the interaction, we conducted 9 pairwise comparisons, similar to those in the analysis for size and color adjectives (corrected using Bonferroni adjustment). In English the comparison between "1" and "3" and the comparison between "2" and "3" revealed significant differences (*ps* < 0.001, see Table A.5 for more details), such that pattern adjective positioned closest to the noun or in the middle position had higher ratings than when positioned farthest from the noun. Naturalness ratings when comparing Distances "2" and "1" in English were not significantly different ($p = 0.030$, not significant after applying Bonferroni correction, significance threshold set at 0.0056). Hebrew speakers displayed the opposite preference: the comparisons between "1" and "3" and between "2" and "3" revealed significant differences (*ps* < 0.001), but with higher ratings for pattern adjectives positioned farthest from the noun. The comparison between Distances "1" and "2" was not significant in Hebrew as well. The comparisons between the languages did not survive correction ($p = 0.055$ for distance "1"; $p = 0.418$ for distance "2"; $p = 0.0127$ for distance "3"). The failure to reach significance could stem from differences in rating patterns between languages (see discussion below).

3.3. Discussion

In this experiment, participants rated phrases with different adjective orders according to their naturalness. Here too we examined the ratings for the order of size and color adjectives in a two-adjective setting and of size, color and pattern in a three-adjective setting. The results from this task support our conclusion following Experiment 1. Like the production task, the naturalness rating task shows a difference in the strength of adjective ordering preferences between English and Hebrew. It also provides further evidence against the claim that the preferred order in Hebrew mirrors that of English.

With regard to two-adjective orders, the results from Experiment 2 (naturalness ratings) are consistent with those of Experiment 1 (production). English speakers rated the Size-Color-Noun order as significantly more natural than the Color-Size-Noun order. In contrast, Hebrew speakers were somewhat indifferent to the relative position of size and color adjectives and did not show a difference in the ratings of the two orders. This is further evidence, using a different method, that English speakers are stricter than Hebrew speakers in their adjective ordering when color and size adjectives are used. Unlike the results from our production experiment, we did not observe any preference in Hebrew

towards the target order (Noun-Color-Size) in this experiment.

In the three-adjective setting, the Size-Color-Pattern-Noun order was rated the most natural in English. The orders rated most natural in Hebrew, Noun-Color-Size-Pattern and Noun-Size-Color-Pattern, were the same orders that were produced most often in Experiment 1. In this experiment, we did not find an effect for language, which could be due to general variance in the ratings: In English, there were orders that were rated quite high (above 6), and orders that were rated quite low (below 4.5). In Hebrew, no order was rated above 5.5 or below 4.5. Thus, on average the languages present similar ratings overall. In any case, the target order (Size-Color-Pattern-Noun) was rated higher in English than the same (mirrored) order in Hebrew. Moreover, none of the orders in Hebrew was rated significantly higher than its English counterpart. In the within language comparison in Hebrew, The Noun-Pattern-Color-Size order was rated significantly lower than the Noun-Color-Size-Pattern order (Table A.4 in Supplementary materials A). This seems to suggest weaker preferences for specific adjective orders in Hebrew. Additionally, these results provide further evidence against the claim that adjective order in postnominal languages mirrors the adjective order in prenominal languages.

We also performed an ad-hoc exploratory analysis, based on the distance of each adjective class from the noun. In English, size was preferred in the farthest position from the noun, whereas the color and pattern adjectives were preferred in the middle or in the closest position to the noun. This finding, like our other results, is in line with theories that argue that "availability" plays the most important role in adjective ordering (Danks & Glucksberg, 1971; Martin, 1969a, 1969b; Whorf, 1945). This is because size, being a gradable adjective, is considered less available than both color and pattern adjective. In Hebrew, however, a different phenomenon was observed: size adjectives were not preferred in a certain position relative to the noun, color adjectives were preferred in the middle position or closest to the noun (like in English), and pattern adjective were preferred the farthest from the noun. This suggests that Hebrew speakers show flexibility in the relative placement of size and color (similar to what is shown in the two-adjective setting). Crucially, this is not predicted by perceptual/conceptual factors alone.

4. Experiment 3: A forced-choice task

In two experiments, we showed that English speakers have stronger preferences than Hebrew speakers do with regard to adjective order when referring to the same visual objects. For nouns modified by 3 adjectives, we also show that the preferred order is different across the languages. In this experiment, we sought to explore the strength of adjective ordering preferences using a forced-choice task. Contrasting two descriptions, differing only with respect to adjective order, and forcing a binary choice between the orders, could potentially reveal stronger preferences in Hebrew than the ones observed when each description was presented individually. Thus, the forced-choice task might help uncover hierarchies between orders that did not present themselves in the previous experiments. Additionally, we did not use pictures in this task, which could help avoid any effects related to visual salience.

4.1. Method

4.1.1. Participants

Sixty individuals participated in this experiment: Thirty were native monolingual Hebrew speakers (20-47 years old, Mean = 28.8) and thirty were native monolingual English speakers (20-44 years old, Mean = 30.43). They had no history of reading difficulties or any other cognitive impairment. Hebrew speakers were recruited via social media, and were paid for their participation. English speakers were recruited via "Prolific". The average time for completing the study was approximately 10 min. Data from two additional English-speaking participants were excluded from analysis, due to their failure in catch trials.

4.1.2. Materials

Thirty-six pairs of descriptions, referring to the same object, were used. Each description included multiple adjectives and a noun, similar to the descriptions used in the previous experiments. The descriptions in each experimental pair differed only with regard to the ordering of the adjectives. For the two-adjective setting, we used six repetitions of the description pairs. For the three-adjective setting, we used thirty description pairs, such that each possible pairing between two different orders repeated twice. These pairs were presented without any picture, to avoid effects of visual salience. To avoid including low quality responses, we used 8 catch trials, in which one of the two descriptions was highly ungrammatical (e.g. "Triangle dotted big green", for English speakers). The other description in the catch trials was always grammatical, but could have been a dispreferred order. We excluded participants who selected the ungrammatical description at least once. The position of each order was counterbalanced (e.g. half of the items in the SC-N/N-CS order were

presented on the left, and half were presented on the right). The order of presentation was randomised across participants.

4.1.3. Procedure

Participants were asked to choose which of the two descriptions in each pair was a more natural description for an imaginary object.

4.2. Results

Like in the previous experiments, we adjusted the adjective ordering (see Table 1).

4.2.1. Two-adjective setting

English speakers chose the SC-N order over the CS-N order in 97.2% of the times, whereas Hebrew speakers made the same (mirrored) selection (N-CS) only in 71.7% of the times (Figure 6a). We conducted a binomial test to examine whether the preferences were above chance-level. We set the chance level at 1/2 because speakers could have chosen one of two possible orders on each trial. As in Experiment 1, both English and Hebrew speakers preferred the SC-N/N-CS order ($ps < 0.001$).

To examine the differences between the two languages, we performed a mixed-effects logistic regression using the lme4 package (Bates et al., 2015) in the R environment (R Core Team, 2015). We modelled the odds of choosing the preferred order in English (SC-N/N-CS), and tested the differences between the two languages. Thus, our fixed effect was language (Hebrew/English). We also included random intercepts for participants and for items (without random slopes due to the failure of the full model to converge). The model revealed an effect of language (Table 9 and Figure 6b), indicating that the SC-N/N-CS order was selected more often in English than in Hebrew.

4.2.2. Three-adjective setting

English speakers showed a clear hierarchy of preferences. The target order, SCP-N, and the SPC-N order were chosen in most of the pairs they appeared in (86% and 73% respectively). All other orders were chosen less than 50% of the time (Figure 7). To examine whether these choices were at chance level, we conducted a binomial test. We set the chance level at 1/2, because in each pair one of two orders could be chosen. This test revealed that the SCP-N and SPC-N orders were reliably selected at above chance level ($ps < 0.001$). Furthermore, the CPS-N, CSP-N and PCS-N orders were reliably chosen below chance level ($ps < 0.01$), whereas the PSC-N order was chosen at chance level ($p = 0.19$).

As in the other two experiments, Hebrew speakers showed a different pattern (Figure 7). The N-CSP and the N-SCP orders were chosen in most of the pairs they appeared in (67% and 59% respectively). Here too we conducted a binomial test. According to this test, the N-CSP and N-SCP orders were reliably chosen at above-chance level ($ps < 0.001$), whereas the N-PSC order was reliably chosen below-chance level ($p < 0.001$). All other orders, N-PCS, N-CPS, and N-SPC were chosen at chance level ($ps > 0.05$). There were only

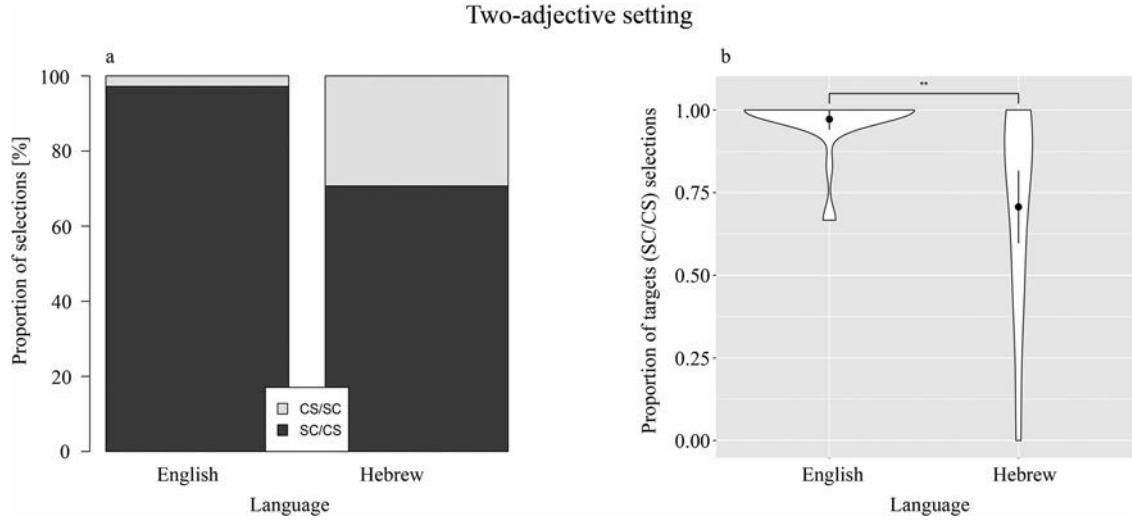

Figure 6. (a) Percentage of the choices of each possible adjective order in the two-adjective setting; (b) Selections of the target order in English (SC-N; N-CS in Hebrew), by language. Points represent group means and error bars represent 95% confidence intervals. S = Size; C = Color. Note that in English (to the left of the slash) the adjectives precede the noun, and in Hebrew (to the right) the noun precedes the adjectives. ** $p < 0.01$.

Table 9. Summary of the mixed-effects logistic regression models for ordering preferences of the target order in two-adjective (SC-N/N-CS) and three-adjective (SCP-N in English and N-CSP in Hebrew[a]) settings in the English (reference) vs Hebrew.

Setting	β	SEM	z-value	*p*
2 adjectives: English vs. Hebrew	7.41	0.002	3953	<0.001***
3 adjectives: English vs. Hebrew	1.61	0.53	3.04	0.002*

[a] In this analysis, we used the most frequent orders in each language, as observed in Experiment 1 and 2, in order to allow a comparison of the strength of the ordering preference.

very few items in each comparison pairs. Even so, we performed further binomial tests for each pair separately. Results for this test are given in Supplementary materials B (Table B.1 for the results from English and Table B.2 for the results from Hebrew). These comparisons revealed a similar pattern to the overall selections with stronger preferences in English than in Hebrew. Specifically, in English there were more pairs than in Hebrew in which the target order was preferred over its competitor.

To examine whether the strength of the preferences in Hebrew and in English differed significantly, a mixed-effects logistic regression analysis was performed, using the lme4 package (Bates et al., 2015) in the R environment (R Core Team, 2015). In this analysis, we used a different target order for each language, based on the results from Experiment 1 and 2. Thus, the target order in English was set to SCP-N, also in accordance with previous accounts, and the target order in Hebrew was set to N-CSP (which was one of the most produced orders in Experiment 1, and the highest rated order in experiment 2). We modelled the odds of choosing the target order, and tested the differences between the two languages. Random intercepts were included for participants and items, as well as random slopes. This analysis revealed a main effect of language (Table 9); such that the corresponding target order was chosen significantly more frequently by English speakers than by Hebrew speakers.

4.3. Discussion

In this experiment, participants selected which of two given multi-adjectivally modified nouns is more natural. The results of this experiment align with the results from the previous ones, suggesting more robust ordering preferences for adjectives in English compared with Hebrew. Critically, these results were obtained in a task without visual stimuli, indicating that the results of Experiments 1 and 2 cannot be explained by visual salience. In the two-adjective setting, the percentage of the Size-Color-Noun/Noun-Color-Size order selection was strikingly similar to the percentage of its productions in Experiment 1, confirming that it is the preferred order in both Hebrew and English. Like in the previous experiments, the strength of this preference was stronger in English.

In the three-adjective setting, like in the previous experiments, Hebrew speakers had a slight preference for the Noun-Color-Size-Pattern and Noun-Size-Color-Pattern orders. Additionally, this experiment further

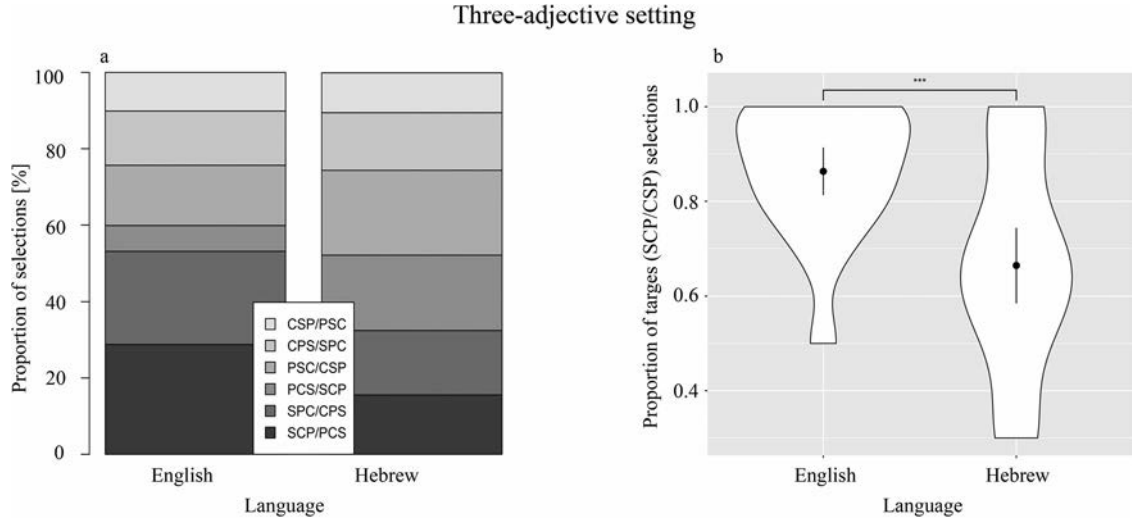

Figure 7. (a) Percentage of choices of each possible adjective order in the three-adjective setting. Note that the preferred order(s) were not present in all pairs, and therefore a certain amount of selection of non-preferred orders is expected. (b) Selections of the target order (SCP-N/N-CSP), by language. Points represent group means and error bars represent 95% confidence intervals. S = Size; C = Color; *P* = Pattern. Note that in English (to the left of the slash) the adjectives precede the noun, and in Hebrew (to the right) the noun precedes the adjectives. *** $p < 0.001$.

confirmed that these preferences were much weaker than the preference for the Size-Color-Pattern-Noun order in English. Again, the results show that the preferred order in Hebrew does not mirror that of English. Because one of the preferred orders in Hebrew is Noun-Size-Color-Pattern, one might argue that the same order of adjective classes is preserved in Hebrew, despite its being a post-nominal language (similar to Irish, see General Discussion). If so, we would expect a strong preference for this order over all other orders. However, this was not the case, as the Noun-Color-Size-pattern order showed stronger preferences (it was reliably preferred in 3/5 possible pairs, whereas Noun-Size-Color-Pattern was reliably preferred in only 1/5, see Table B.2 in supplementary materials B). This result, in which the preferred order in Hebrew neither mirrors nor follows the order in English, cannot be explained solely by universal conceptual/perceptual constraints.

5. Experiment 4: A forced-choice task with several semantic classes

In the experiments reported above, we showed converging evidence, under three different task demands, that adjective ordering preferences in English speakers and Hebrew speakers are different, in both strength and nature. However, in those experiments, we only used adjectives from three specific semantic classes: size, color and pattern. To be able to generalise this conclusion, we must show that it occurs with other semantic classes as well. In this experiment, we used adjectives from six classes: Age, Color, Opinion, Shape, Size and Texture. Note that the size class included both size (big, small) and length (short, long) words, thus expanding on words in this class as well. To examine ordering of these classes, we had to use a forced-choice task, which did not require visualisation. Due to the larger number of semantic classes, we used only the two-adjective settings. Finally, we also used real-world objects (table, chair etc.), to make the experimental setting more natural.

5.1. Method

5.1.1. Participants

Sixty individuals participated in this experiment: Thirty were native monolingual Hebrew speakers (20-35 years old, Mean = 25.9) and thirty were native monolingual English speakers (20-40 years old, Mean = 28.3). They had no history of reading difficulties or any other cognitive impairment. Hebrew speakers were recruited via social media, and were paid for their participation. English speakers were recruited via "Prolific". The average time for completing the study was approximately 10 min. Data from one additional English-speaking participant were excluded from analysis, due to their failure in catch trials.

5.1.2. Materials

Twenty-four pairs of descriptions, referring to the same object, were used. In this experiment, we included "real-world" nouns, instead of shapes. The pairs differed only in the ordering of the adjectives. Similarly to Experiment 3, these pairs were presented without any picture. The semantic classes we used in this experiment were Age (modern, new, old), Color (blue, green, red, yellow), Opinion (beautiful, ugly), Shape (round, square), Size (big, long, short, small) and Texture (hard, smooth, soft). Nouns were real-world objects (carpet, chair, curtain, mattress, sink, table). All adjectives/noun combinations were plausible (e.g. "an ugly soft carpet", but not "an ugly soft sink"). The experimental pairs were Age-Color, Age-Shape, Opinion-Texture and Size-Texture. Each pairing repeated 6 times, as in Experiment 3. To avoid including low quality responses, we included 6 catch trials, similar to those used in Experiment 3 (i.e. one of the two descriptions was highly ungrammatical, such as "a sink new green"). We excluded participants that selected the ungrammatical order at least once. The position of each order was counterbalanced (e.g. half of the items in the Age-Color-Noun order were presented on the left, and half were presented on the right). The order of presentation was randomised across participants.

5.1.3. Procedure

The same procedure as in Experiment 3 was used.

5.2. Results

In the following section, we report the results of each pair separately. The same analyses were conducted, similar to Experiment 3. First, we conducted a binomial test to examine whether the preferences of were above chance-level. We set the chance level at 1/2 because speakers could have chosen one of two possible orders on each trial. Then, we examined the differences between the two languages, using a mixed-effects logistic regression (the lme4 package) (Bates et al., 2015) in the R environment (R Core Team, 2015). In each pair, we modelled the odds of choosing of the preferred order in English with a fixed effect of language (Hebrew/English). We also included random intercepts and random slopes for participants and for items (the statistical results are given in Table 10).

Table 10. Summary of the mixed-effects logistic regression models for ordering preferences of the target orders (AC-N/N-CA, Ash-N/N-ShA, OT-N/N-TO, ST-N/N-TS) in English (reference) vs. Hebrew.

Class pair	β	SE	z	*p*
Age-Color: English vs. Hebrew	−2.16	0.43	−5.01	<0.001***
Age-Shape: English vs. Hebrew	−1.35	0.36	−3.80	<0.001***
Opinion-Texture: English vs. Hebrew	−0.12	0.33	−0.37	0.715
Size-Texture: English vs. Hebrew	−1.90	0.26	−7.40	<0.001***

5.2.1. Age & Color items

English speakers chose the Age-Color-Noun (AC-N) order over the Color-Age-Noun (CA-N) order in 95% of the times, whereas Hebrew speakers made the same selection (i.e. N-CA/Noun-Color-Age) only in 65.6% of the times (Figure 8a). The binomial test showed that both English and Hebrew speakers preferred the AC-N/N-CA order at above chance level (*ps* < 0.001). The mixed-effects model revealed an effect of language (Table 10), indicating that the AC-N order was selected more often in English than the N-CA order in Hebrew (Figure 8b).

5.2.2. Age & Shape items

English speakers chose the Age-Shape-Noun (Ash-N) order over the Shape-Age-Noun (ShA-N) order in 86.7% of the times, whereas Hebrew speakers made the same selection (i.e. N-ShA/Noun-Shape-Age) only in 64.4% of the times (Figure 9a). The binomial test showed that both English and Hebrew speakers preferred the ASh-N/N-ShA order at above chance level (*ps* < 0.001). The mixed-effects model revealed an effect of language (Table 10), indicating that the Ash-N/N-ShA order was selected more often in English than in Hebrew (Figure 9b).

5.2.3. Opinion & Texture items

English speakers chose the Opinion-Texture-Noun (OT-N) order over the Texture-Opinion-Noun (TO-N) order in 65.6% of the times, and Hebrew speakers made the same selection (i.e. N-TO/Noun-Texture-Opinion) in 63.9% of the times (Figure 10a). The binomial test showed that both English and Hebrew speakers preferred the OT-N/N-TO order at above chance level (*ps* < 0.001). The mixed effects model revealed no significant effect of language (Table 10; Figure 10b).

5.2.4. Size & Texture items

English speakers chose the Size-Texture-Noun (ST-N) order over the Texture-Size-Noun (TS-N) order in 79.4% of the times, whereas Hebrew speakers made the same selection (i.e. N-TS/Noun-Texture-Size) only in 37.2% of the times (Figure 11a). The binomial test showed that English speakers preferred the ST-N order at above chance level. However, Hebrew speakers preferred the opposite order, N-ST (Noun-Size-Texture), at above chance level (*ps* < 0.001). The mixed-effects model revealed an effect of language (Table 10), indicating that the ST-N/N-TS order was selected more often in English than in Hebrew.

To examine whether the preferences in Hebrew differ from the ones in English not only in nature, but also in strength, we conducted an ad-hoc comparison. In this

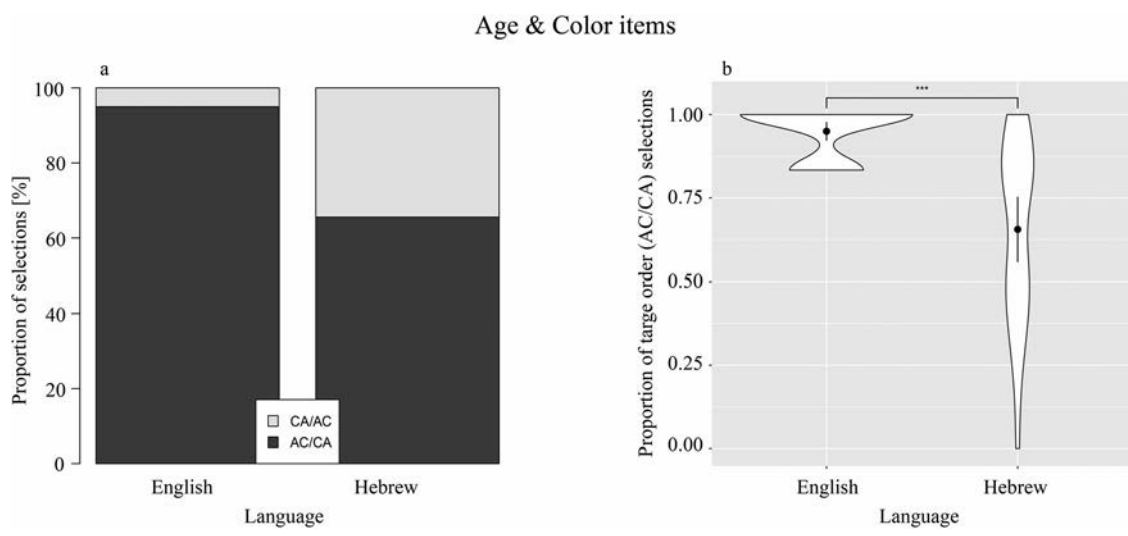

Figure 8. (a) Percentage of the choices of each possible adjective order in the age-color items; (b) Selections of the target order in English (AC-N; N-CA in Hebrew), by language. Points represent group means and error bars represent 95% confidence intervals. A = Age; C = Color. Note that in English (to the left of the slash) the adjectives precede the noun, and in Hebrew (to the right) the noun precedes the adjectives. *** $p < 0.001$.

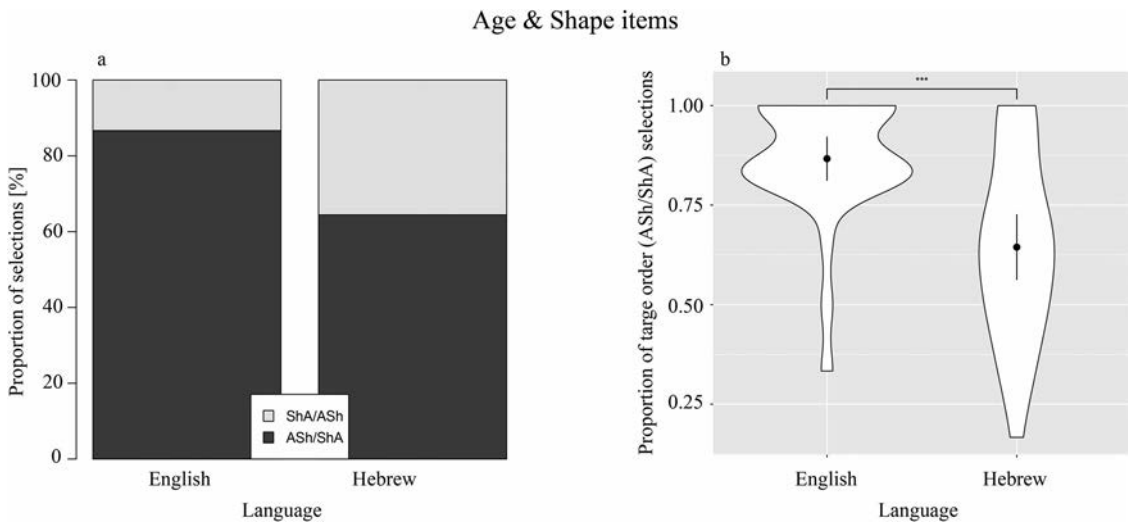

Figure 9. (a) Percentage of the choices of each possible adjective order in the age-shape items; (b) Selections of the target order in English (Ash-N; N-ShA in Hebrew), by language. Points represent group means and error bars represent 95% confidence intervals. A = Age; Sh = Shape. Note that in English (to the left of the slash) the adjectives precede the noun, and in Hebrew (to the right) the noun precedes the adjectives. *** $p < 0.001$.

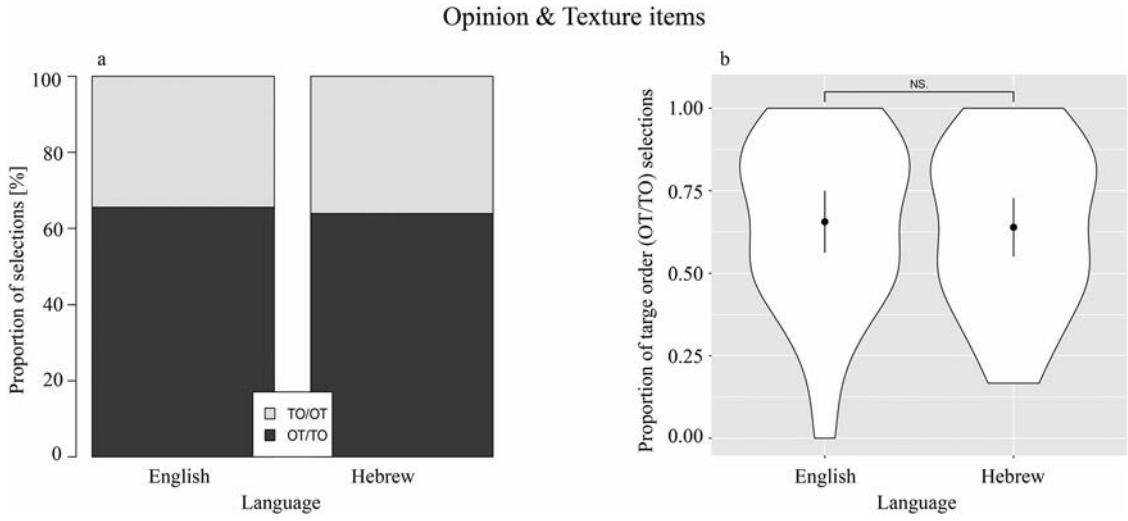

Figure 10. (a) Percentage of the choices of each possible adjective order in the opinion-texture items; (b) Selections of the target order in English (OT-N; N-TO in Hebrew), by language. Points represent group means and error bars represent 95% confidence intervals. O = Opinion; T = Texture. Note that in English (to the left of the slash) the adjectives precede the noun, and in Hebrew (to the right) the noun precedes the adjectives. NS. = non-significant

analysis, we modelled the odds of choosing of the preferred order in each language with a fixed effect of language (Hebrew/English), using a different order in each language (ST-N for English and N-ST for Hebrew). We also included random intercepts and random slopes for participants and for items. This model revealed a main effect of language ($p = 0.0017$; Table 11), such that the strength of preference in English was stronger than the one in Hebrew. Thus, even when taking into account the preferred order in Hebrew, English speakers show more robust ordering preferences.

5.3. Discussion

This experiment aimed to generalise the findings from Experiments 1-3, which included adjectives from three semantic classes only. In this experiment, we included six semantic classes in a forced-choice task. The results support the results of the previous experiments, showing more robust preferences in English for some multi-adjective phrases, as well as different ordering preferences for others.

The four adjective combinations examined revealed mixed results with regard to adjective ordering

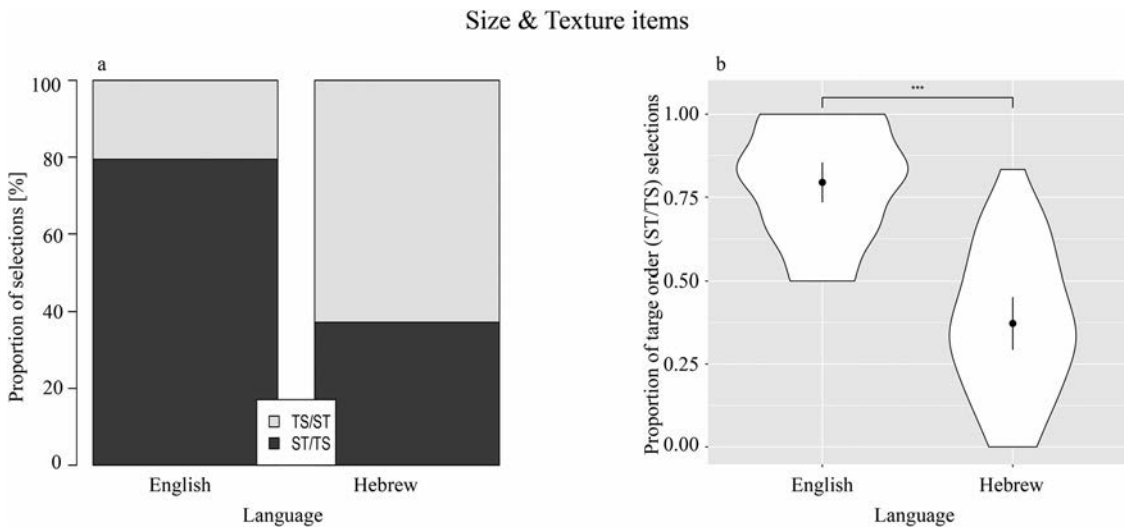

Figure 11. (a) Percentage of the choices of each possible adjective order in the opinion-texture items; (b) Selections of the target order in English (ST-N; N-TS in Hebrew), by language. Points represent group means and error bars represent 95% confidence intervals. S = Size; T = Texture. Note that in English (to the left of the slash) the adjectives precede the noun, and in Hebrew (to the right) the noun precedes the adjectives. *** $p < 0.001$

preferences in English compared with Hebrew. In three combinations, the same ordering was preferred, with the Hebrew order mirroring that of English. Of these three, one combination (Opinion-Texture) showed no difference between the languages. Note that a rather weak preference (less than 70%) for the Opinion-Texture-Noun order was observed even in English. In the other two combinations (Age-Color, Age-Shape), the preference in Hebrew was significantly weaker than in English. This result is similar to the one observed for the size-color combination tested in the previous experiments.

In the fourth combination, that of size and texture adjectives, the preferred order in Hebrew did not mirror that of English. Furthermore, the preferred order in Hebrew was preferred less strongly than in English. This result provides further evidence that adjective ordering preferences in Hebrew does not necessarily mirror the preferences in English, and that Hebrew consistently displays weaker preferences for specific adjective orders. This is in line with the results from the previous experiments which showed that, in the three-adjective setting with the classes of size, color and pattern, the preferred adjective ordering in Hebrew did not mirror that of English. In this experiment, we showed that this is evident with other classes, as well as when only two adjectives are included.

Table 11. Summary of the mixed-effects logistic regression model for ordering preferences of the preferred orders (ST-N/N-ST) in English (reference) vs. Hebrew, when using size and texture adjectives.

Class pair	β	SE	Z	*p*
Size - Texture (preferred orders): English vs. Hebrew	−0.81	0.26	−3.143	0.00167**

6 General discussion

This study examined adjective ordering preferences within multi-adjectival phrases in two languages: Hebrew, a postnominal language and English, a prenominal language. Three experiments focussed on a systematic investigation of three specific adjective classes (size, color and pattern) using three different experimental tasks: a production task, a naturalness ratings task and a forced-choice task. A fourth experiment was aimed to generalise our findings to additional semantic classes of adjectives and adjective combinations (age, shape, texture, and opinion), using a forced choice task. In the three tasks testing size-color combinations, English speakers showed a clear preference to use the Size-Color-Noun order. Hebrew speakers preferred the Noun-Color-Size order in the production and forced-choice tasks. This order mirrors the preferred order in English, which is expected according to several accounts (Shlonsky, 2004; Sproat & Shih, 1991). This mirrored ordering was also observed with some combinations (age-color and age-shape) in Experiment 4. Critically, the ordering preference in these cases was significantly weaker in Hebrew compared with English.

English speakers presented a clear preference also with three adjectives in Experiments 1–3 (albeit slightly weaker than the one observed with two adjectives): they favoured the Size-Color-Pattern-Noun order, and

accepted Size-Pattern-Color-Noun as a decent alternative. In contrast, Hebrew speakers did not have such clear preferences. They showed a slight tendency for the Noun-Size-Color-Pattern and Noun-Color-Size-Pattern orders. Note that in this case, both orders do not mirror the preferred order in English. Whereas one follows the same linear adjective order (with the noun positioned before the adjective in Hebrew), the other bares no direct relation to the preferred English order. Preference for the same linear adjective order emerged not only in the three-adjective setting, but also when using two-adjective setting with size and texture adjectives. English speakers consistently preferred the Size-Texture-Noun order, while Hebrew speakers preferred (more weakly) the Noun-Size-Texture order.

These results raise the question what determines adjective ordering across languages. Several accounts have attempted to form a universal explanation of adjective ordering preferences. Theoretical/observational accounts have mostly focussed on semantic/grammatical constraints, such that the semantic class of the adjective determines its structural relation to the noun it modifies (Cinque, 1994; Dixon, 1982; Laenzlinger, 2005; Scott, 2002; Svenonius, 2008). These accounts offer that various hierarchies between the different adjective classes rigidly determine their ordering within any multi-adjectival phrase. Different hierarchies have been assumed by different researchers, and most of them include size, color, age, shape and quality (or opinion/evaluative comment), such that the hierarchy is as follows: opinion > size > age > shape > color. The higher an adjective is in this hierarchy, the farther it is expected to be from the noun it modifies, when multiple-adjectives are present. This means, for example, that size adjectives are expected to be farther from the noun than color adjectives.

Still, some have tried to account for cross-linguistic variation in adjective ordering preferences (Cinque, 1994; Sproat & Shih, 1991). For example, Cinque (1994) offers a grammatical account where variations are achieved by means of movement within the syntactic tree. According to this account, the linear order of the adjectives stays the same in post-nominal languages, but the noun "moves" towards the beginning of the phrase and crosses some of the adjectives, depending on their semantic classes. Such an account could potentially explain the differences between the languages. Sproat and Shih (1991) argue that different languages have different adjectival modification which can account for the variability of adjective ordering. Languages with parallel modification (where adjectives modify the noun in a parallel, and not hierarchical, manner), and languages with indirect modification (such as modification by relative clauses; e.g. "a circle which is blue") do not display adjective ordering restrictions. However, Shlonsky (2004) offers a grammatical description of Hebrew based on Cinque's account. He suggests that adjective ordering preferences in Hebrew mirror the ones in English. Furthermore, he argues that Hebrew has a hierarchical (or in Sproat & Shih's terminology - direct) modification. Therefore, these accounts cannot explain the variation between languages that was observed in the present study.

Other accounts have focussed on the perceptual/conceptual properties of adjectives (Belke, 2006; Danks & Glucksberg, 1971; Martin, 1969a, 1969b; Scontras et al., 2017, 2019), also trying to explain the hierarchies proposed by the above accounts. These accounts assume that perceptual or conceptual availability determines the ordering of multi adjectival phrases. Availability has been given different definitions, and may involve absoluteness (Martin, 1969a) or subjectivity (Scontras et al., 2017, 2019). No matter the type of definition, it is argued that the more available an adjective is, the closer it will be to the noun it modifies. For example, adjectives that denote size require a reference point (i.e. to determine whether a circle is big or small, other items are needed) unlike color or pattern. That is, color and pattern are considered absolute and less subjective. Therefore, like in the grammatical accounts, size adjectives are expected to be farther from the noun than color and pattern adjectives.

This prediction was indeed maintained in our study when testing English with size, color and pattern. English speakers consistently preferred the Size-Color-Noun order in the two-adjective setting, and the Size-Color-Pattern-Noun order in the three-adjective setting. This was also observed in the two-adjective setting in Hebrew, where color adjectives were preferred at positions closer to the noun than size adjectives in the production and selection tasks (although no such preference was observed in Experiment 2, see more discussion below). This extends to other classes tested in Experiment 4. Both English and Hebrew speakers preferred age adjectives in the farther position from the noun, when testing age-color and age-shape pairs.

However, this was not the case in the three-adjective setting, and in one of the pairs of classes in Experiment 4 (Size-Texture). Hebrew speakers consistently preferred to place pattern adjectives at the position farthest from the noun: both the Noun-Color-Size-Pattern and the Noun-Size-Color-Pattern orders were produced and selected more often than the other orders, as well as rated higher. This was also supported by the exploratory analysis following the naturalness rating task (Experiment 2), which showed that beyond the specific

order, pattern adjectives were rated highest in Hebrew when they were in the position farthest from the noun. Pattern adjectives are considered more available than size adjectives, because they are absolute and less subjective.

A similar finding was observed in Experiment 4, with size and texture adjectives. In English, as expected based on previous grammatical and conceptual accounts (Scontras et al., 2017; Svenonius, 2008), texture adjectives were preferred closer to the noun when paired with size adjectives. In the same pairing in Hebrew size adjectives were preferred at closer proximity to the noun. Size is considered more subjective than texture (Scontras et al., 2017, 2020), thus, under conceptual accounts, it should be placed farther from the noun. This is not the case in Hebrew. Note that texture adjectives form a different semantic class than pattern adjectives.[3] Therefore, the differences between Hebrew and English can be generalised beyond a specific class of adjectives.

The finding stated above - that less subjective adjectives in Hebrew, from two different semantic classes, were not preferred at positions in close proximity to the noun - is incompatible with the conceptual accounts. To explain our results, we turn to accounts that suggest that several properties of the adjectives, not solely conceptual ones, influence the ordering of adjectives in multi-adjectival phrase (Trotzke & Wittenberg, 2019; Wulff, 2003; Wulff & Gries, 2015). These could be semantic or conceptual factors (e.g. availability or absoluteness) as suggested by several accounts, but also lexical (e.g. length or frequency; see also Scontras et al., 2017, for an examination of these factors), or phonological (e.g. syllable structure) factors. These factors interact when determining the ordering of the adjectives. Corpus analyses and experimental results have shown that length and frequency contribute to the ordering preferences in English (Scontras et al., 2017; Trotzke & Wittenberg, 2019; Wulff, 2003; Wulff & Gries, 2015). Specifically, shorter adjectives tend to precede longer ones, and more frequent adjectives also tend to precede less frequent ones.

Let us consider the three-adjective setting as an example. In English, the conceptual factors and lexical factors are assumed to operate in the same direction. Based on the conceptual factors, the pattern adjectives, which are more absolute/less subjective, should be in the closest proximity to the noun. Based on the lexical factors, our pattern adjectives, which are longer and less frequent than the size and color adjectives (see supplementary materials), should appear later in the sequence. Thus, both conceptual and lexical factors predict, independently, that our pattern adjectives will appear after the size and color adjectives.

Applying the same logic to the Hebrew data will result in a conflict. The conceptual factors predict the same ordering as English, such that pattern adjectives should be placed in close proximity to the noun. Our pattern adjectives were longer and less frequent than both size and color adjectives (see Supplementary materials C). Thus, the lexical factors motivate placing the pattern adjectives later in the sequence. Because Hebrew is a post-nominal language, later in the sequence means farther from the noun. Thus, the placement of pattern adjectives is subject to two different, opposite, forces. Note that this is true only if longer and infrequent adjectives follow shorter and frequent adjectives, regardless of the noun position with respect to its adjectives. That is, the conflict between conceptual and lexical factors is expected only if length and frequency effects are to be applied linearly, and not as a function of distance from the noun (see Vandekerckhove et al. (2015) for an account on linear determinants of adjective order). Thus, accounts that allow for several influences on adjective ordering could explain the results from our study better than purely conceptual ones. These accounts do not describe what happens when a conflict between two or more factors occurs (e.g. in cases where conceptual factors promote proximity to the noun, but lexical factors promote placement in earlier position in prenominal languages or in later positions in postnominal languages). It is also an open question whether these factors operate differently across languages. Further research investigating the effects of both lexical and conceptual (and possibly phonological) factors on adjective ordering preferences could help answer this question (especially in postnominal languages, where empirical data is sparse).

In addition to the differences between Hebrew and English in the ordering of the adjective classes, we also need to consider the differences in the robustness of the preference of speakers of those languages. English speakers had clear, more robust, preferences than Hebrew speakers. A possible explanation for this difference is in the pragmatic needs that arise from the different positions of the noun with regard to its adjectives. It has been suggested that adjectives play a different role in prenominal and postnominal languages. In prenominal languages, adjectives appear before the noun and therefore, speakers of such languages can rely on the adjectives in narrowing down the number of possible referents (Engelhardt et al., 2008), even when the adjectives are redundant (Rubio-Fernández, 2016). This is not the case for speakers of postnominal languages, where narrowing down based on adjectives is felicitous only when they are not redundant (although see Rubio-Fernández, 2016, for complex settings in

postnominal languages). If so, adjectives seem to be more critical for object identification in prenominal languages than in postnominal languages (Franke et al., 2019; Rubio-Fernández, 2016, 2019). Indeed, it is well-established that English speakers often produce color adjectives in an array of unique objects (i.e. when the color is unneeded for identification; Koolen et al., 2013; Rubio-Fernández, 2016; Tarenskeen et al., 2015). Spanish speakers, for which the adjective appears after the noun, do so only under special circumstances (i.e. in very large arrays, Rubio-Fernández, 2016, 2019). If adjective use is more effective in prenominal languages, it could be helpful for these languages to develop a uniform adjective ordering to reduce processing demands. In contrast, postnominal languages may allow a flexible adjective ordering, resulting in the pattern of linguistic behaviour observed in Hebrew.

Another post-nominal language shows flexible ordering of multi-adjective phrases. Recent empirical evidence suggests that in Spanish, adjective ordering constraints are virtually absent (Rosales Jr & Scontras, 2019). However, a conjunction structure was used (e.g. a big *and* red car). Conjunction is claimed to "delete" ordering preferences in English (Byrne, 1979; Ford & Olson, 1975). In fact, Sproat and Shih (1991) argued that conjunction between adjectives is preferred, in some cases even mandatory, in languages with parallel modification which have flexible adjective ordering. Rosales and Scontras concluded that Spanish lacks adjective ordering preferences because it always requires conjunction in multiple-adjective strings.

In our production experiment (Experiment 1), most productions did not include conjunction (89% in the two-adjective setting and 82% in the three-adjective setting). This clearly indicates that conjunction is not preferred or required in multiple-adjective strings in Hebrew. Thus, the weaker ordering preferences in Hebrew, and the different preferred orders cannot be accounted for by the explanation suggested for Spanish.

One of the aims of our study was to determine whether mirrored adjective ordering is preferred in postnominal languages. This claim is not supported by our data from the three-adjective setting and from size and texture adjectives. It is worth noting that some postnominal languages, such as Irish, are reported to have the same linear order as prenominal languages with the noun appearing at the beginning of the phrase. According to this, an Irish speaker should use the equivalent of "a cup large green" (in Irish, "cupán múr uaine"; Laenzlinger, 2005). Indeed, one of the preferred orders in Hebrew, in the three-adjective setting, was the equivalent of linear preferred order in English (noun-size-color-pattern, coded as PCS-N/N-SCP, see Table 1). However, if Hebrew speakers would have displayed a similar pattern of preferences to Irish, we would expect this order to be reliably preferred over all other orders. This is not what the data suggests, because in Experiments 1-3 this order is not reliably preferred over the Noun-Color-Size-Pattern order (coded as PSC-N/N-CSP). Moreover, in most two-adjective settings, the order in Hebrew which mirrors the preferred order in English was slightly preferred over the same linear order. Thus, we cannot argue that the ordering preference in Hebrew mirrors nor mimics the linear ordering preferences in English.

Finally, we would like to draw the readers' attention to the uniqueness of the naturalness task (Experiment 2; see also Leivada & Westergaard, 2019 for a similar task in Greek). In this task, no significant difference was observed between the ratings of the two possible orders in Hebrew in the two-adjective setting. Because participants rated the orders independently from the other orders, they could have given similar rates to several orders. This was not the case for the other two tasks where only one response was allowed. In Experiment 1, participants had to produce only one of several alternative orders, even if they considered other orders to be acceptable or natural. In Experiments 3 & 4, participants had to explicitly choose one of two possible orders, and again could have also found the other (unselected) order acceptable. Thus, the naturalness task was the only task in which participants were *not* required to select a single order. This finding, that noun-color-size and noun-size-color orders are both considered natural in Hebrew (especially when considering the lower ratings of the items in the three-adjective setting), suggests that both orders are acceptable in Hebrew despite the preference for the noun-color-size order. As so, this further emphasizes the differences between Hebrew and English with regard to ordering preferences.

To conclude, we have shown that Hebrew speakers do not only have weaker adjective ordering in multi-adjectival phrase, but they also sometimes prefer completely different orders. This was observed under different task demands, and with several adjective classes. Overall, our data could be better explained by taking into account not only semantic/conceptual factors to explain adjective ordering preferences, but also lexical properties of the adjectives such as frequency and length.

Notes

1. Productions with conjunctions were too few to analyze, and most were produced by the same participants.

Nevertheless, descriptively, it seems that the conjunction 'deleted' the preference for the N-SC order over the N-CS order in the two-adjective setting. In the three-adjective setting, all the productions with conjunctions were in the N-CSP or N-SCP orders.

2. The considerable number of excluded English-speaking participants was because we were unfamiliar with the option to prescreen based on low quality participation in previous studies. In a task like naturalness-ratings, participants can provide low effort responses, which were, in this case, uniform ratings in all trials, as well as giving high ratings to our catch trials, which included incorrect descriptions of the image (e.g., accepting the description "a big blue star" for a picture of a big red star).
3. A few obvious differences are that texture adjectives are gradable (can appear with almost, e.g., "almost smooth"), but pattern adjectives are more objective (*almost dotted, also see the faultless disagreement task, Scontras et al., 2017, 2020). Additionally, texture adjectives, unlike pattern adjective, cannot be used in prepositional phrases (e.g., *"a pillow with smoothness", but "a pillow with dots"). Finally, texture adjectives are not denominal (at least in Hebrew and English), whereas pattern adjectives are.

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